RapidSims | Adding New Simulations

Deployment runbook • OneSmarterInc/RapidSims • October 2026

Use this guide whenever a new simulation must be hosted on rapidsims.flexee.org. The current workflow deploys tested commits from master to EC2, provisions registered systemd services and generates Nginx routes.

1. Before you begin

- Create the new simulation code in a lowercase folder name, e.g. sim11 or sim-career. Folder names must match [a-z0-9-]+; uppercase letters and underscores are rejected by ec2/render-nginx.js.
- Ensure the new application has a package.json in its root and that its build command, if present, succeeds. The EC2 adapter serves index.html or public/index.html and supports api/*.js routes.
- Choose a unique service ID, port and URL prefix; do not overwrite existing entries in ec2/apps.json. Check compatibility with the shared Node runtime and platform authentication/reporting requirements.

2. Create the app folder

Example for a new RapidSim 11:

```
sim11/
  package.json
  index.html   (or public/index.html)
  api/         (optional Vercel-style handlers)
  other game assets / build inputs
```

Minimal package.json:

```
{
  "name": "sim11",
  "private": true,
  "scripts": { "test": "node -e \"process.exit(0)\"" }
}
```

The example is a hosting smoke test, not a complete student simulation. Production games require their own gameplay, API and appropriate security tests.

3. Register the simulation in ec2/apps.json

Add an entry within the existing JSON object (include a comma between adjacent entries):

```
"sim11": {
  "dir": "sim11",
  "port": 3114,
```

```
  "prefix": "/sim11",
  "title": "rapid-sim-11"
}
```

Port 3114 is an example following the simSahil test on 3113; verify availability before using it. The ID becomes rapidsims@sim11.service. The prefix becomes https://rapidsims.flexee.org/sim11/.

4. Commit and deploy

Commit both the new directory and ec2/apps.json, then push to master. The existing GitHub Actions workflow (.github/workflows/deploy-ec2.yml) runs npm test and deploys the exact verified commit on the self-hosted EC2 runner.

```
git switch master
git pull --ff-only origin master
git add sim11 ec2/apps.json
git commit -m "Add RapidSim 11"
git push origin master
```

Observe the deployment under: https://github.com/OneSmarterInc/RapidSims/actions/workflows/deploy-ec2.yml

5. What the CD system does

Stage	Automatic action
CI	Install dependencies; run repository tests and shell syntax checks.
Release	Fetch tested SHA, build app assets and apply existing database migration checks.
systemd	Update per-app release link; enable and restart rapidsims@sim11.service via the shared template.
Nginx	Generate /sim11/ route, validate nginx -t and reload Nginx.
Verification	Run authenticated service readiness checks. On failure, deployment attempts rollback.

6. Verify on EC2

```
sudo systemctl status rapidsims@sim11.service --no-pager
sudo systemctl is-enabled rapidsims@sim11.service
curl -I http://127.0.0.1:3114/
curl -I https://rapidsims.flexee.org/sim11/
sudo nginx -t
```

sudo systemctl list-units "rapidsims@*.service" --all
Expected: active (running), enabled, HTTP 200 for a page that exists, and a successful Nginx config test. Finally, open the public URL and perform an actual student/instructor walkthrough.

7. Troubleshooting

CI fails: Invalid app manifest

Check app directory capitalization, route prefixes and unique numeric ports. Example: simSahil was rejected; sim-sahil is valid.

CI fails in another test

Keep the GitHub Actions run and inspect the failing test lines; fix code or expectations rather than bypassing CI.

Deployment fails on EC2

Review the Deploy production job logs and sudo journalctl -u rapidsims@sim11.service -n 100 --no-pager. Check environment overrides and readiness dependencies.

Service exists but URL returns 404

Check ec2/apps.json registration, successful deployment, generated Nginx route and the presence of index.html/public/index.html.

Do not manually repair first

Leave existing services and the shared rapidsims@.service template intact; diagnose failed deployment and rerun after a corrected push.

8. Hosting is not catalogue integration

The new systemd service and URL do not automatically add the game to the faculty/student catalogue, enrolment, launch-token configuration or results/report workflow. Those are separate application-level integration tasks and should be tested before announcing the simulation as available to students.

Quick release checklist

☐ Lowercase app folder and valid package.json

☐ ec2/apps.json entry uses unique ID, port and prefix

☐ Tests pass and push is on master

☐ Verify and Deploy production jobs green

☐ Service active/enabled; local and public URL reachable

☐ Student/faculty experience and results integration tested (if applicable)